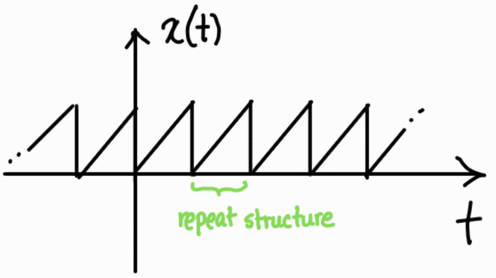
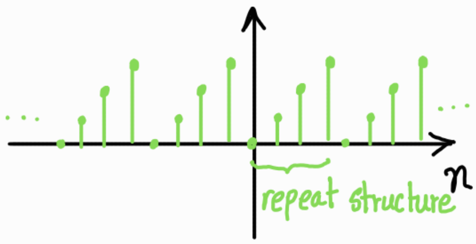
From here on out we will focus completely on the frequency domain. Before we go to Fourier, we need some additional prelims which we will cover in this lecture.

① Periodic signals

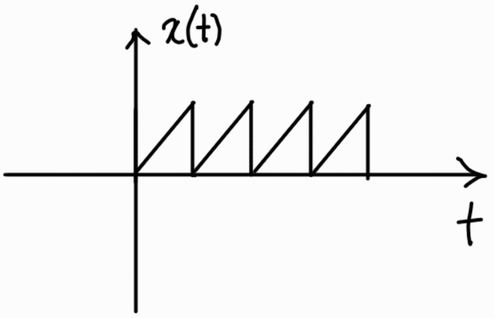
Periodic signals have a repititive structure that is repeated from $-\infty$ to $+\infty$.



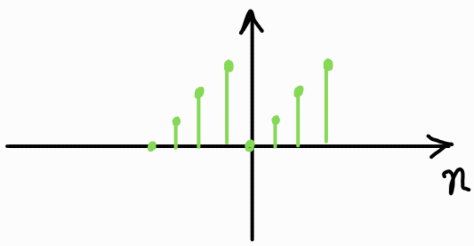
CT periodic signal



DT periodic signal



CT aperiodic signal
doesn't repeat from $-\infty$ to $+\infty$

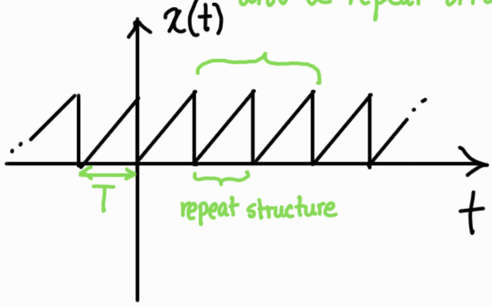


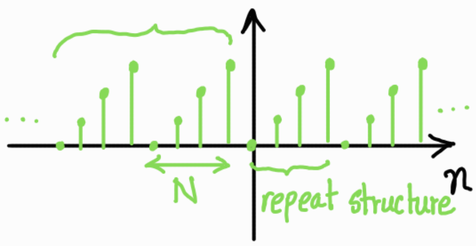
DT aperiodic signal
doesn't repeat from $-\infty$ to $+\infty$

Some defining characteristics of periodic signals —

* The time/n (samples) taken for the shortest repeat structure is called the fundamental time period, represented with T for CT signals and N for DT signals.

also a repeat structure but not the shortest





* Frequency in a CT signal represents the number of oscillation per second and is given by $f = \frac{1}{T}$. The unit of frequency is in Hz. Unlike CT signals, frequency in DT signals are defined differently. In fact there are a total of 3 frequencies in play for DT signals —

1. frequency of the original CT signal $\longrightarrow f = \frac{1}{T}$
2. Sampling frequency $\longrightarrow F_S = \frac{1}{T_S}$
3. Normalized frequency $\longrightarrow f_{norm} = \frac{f}{F_S}$ [ranges from $-\frac{1}{2}$ to $\frac{1}{2}$ or 0 to 1 depending on representation]

don't worry if you don't get it now

We will learn about it more using discrete sinusoids as case study

* $x(t) = x(t \pm k \cdot T)$ & $x[n] = x[n \pm k \cdot N]$ where $k \in \mathbb{Z}$

CT sinusoids —

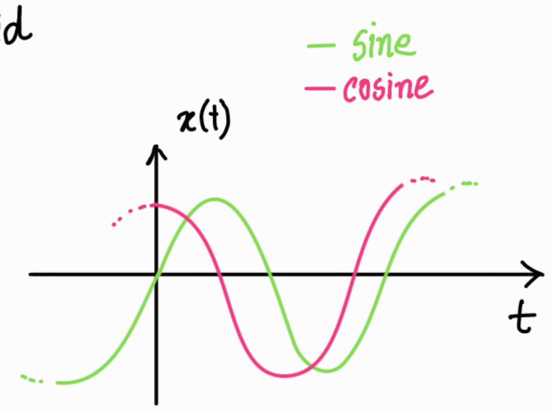
$x(t) = A\sin(2\pi f t - \phi)$ is the standard sinusoid

$x(t) = A\cos(2\pi f t - \phi)$ is the cosine variant

Here, A is the amplitude

f is the frequency

ϕ is the initial phase



for sinusoids we define a different frequency, angular frequency, $\omega = 2\pi f$

do note that $\int_0^T A\sin(2\pi f t) \cdot A\cos(2\pi f t)\, dt = 0 \leftarrow$ orthogonality

CT sinusoids are always periodic.

Proof that $x(t) = x(t \pm KT)$ for CT sinusoids —

Let $x(t) = A\sin(2\pi f t - \phi)$

then, $x(t \pm KT) = A\sin(2\pi f t \pm 2\pi f KT - \phi)$

we know, $f = \frac{1}{T} \rightarrow fT = 1$

$\therefore x(t \pm KT) = A\sin(\pm 2k\pi + 2\pi f t + \phi) = A\sin(2\pi f t + \phi)$ since $k \in \mathbb{Z}$

DT sinusoids —

DT sinusoids are defined as $x[n] = A\sin(\Omega n - \phi)$

or $x[n] = A\cos(\Omega n - \phi)$

$\hookrightarrow$ you can use ω_0 instead

How did we get here?

Let $x(t) = A\sin(2\pi f t - \phi)$ be our desired CT sinusoid

Now we sample $x(t)$ with a sampling duration of T_s —

$$x(nT_s) = x[n] = A\sin(2\pi f \cdot nT_s - \phi)$$

$$= A\sin\left(2\pi \frac{f}{F_s} \cdot n - \phi\right) \quad \left[\because T_s = \frac{1}{F_s}\right]$$

$$= A\sin(2\pi f_{norm} n - \phi)$$

$$= A\sin(\Omega n - \phi)$$, normalized angular frequency, $\Omega = 2\pi f_{norm}$

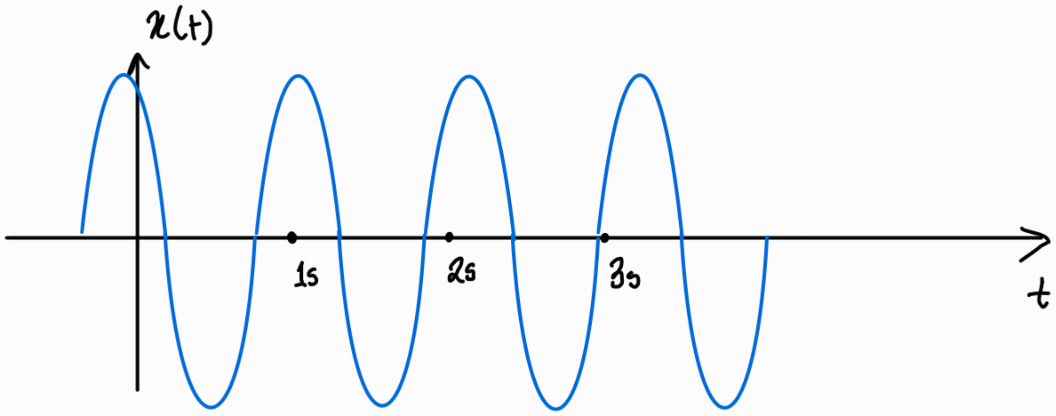
$f_{norm} \longrightarrow$ cycles/sample

$\Omega \longrightarrow$ rad/sample

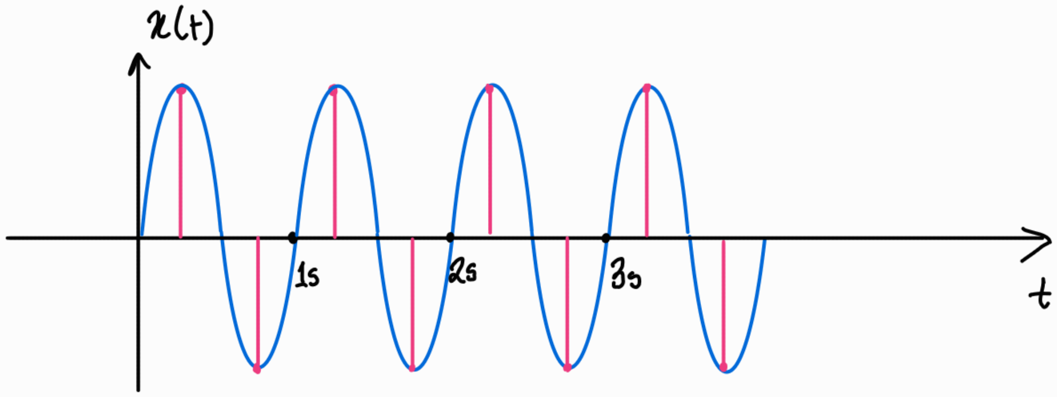
Nyquist sampling theorem & a deep dive into normalized frequency —

Nyquist sampling theorem states that to accurately convert an analog signal to a digital signal and avoid aliasing, the sampling frequency must be double the max. frequency in the signal.

This is kind of complex. So lets start slow and we will understand the whole concept eventually. Lets take a 1 Hz sine wave —

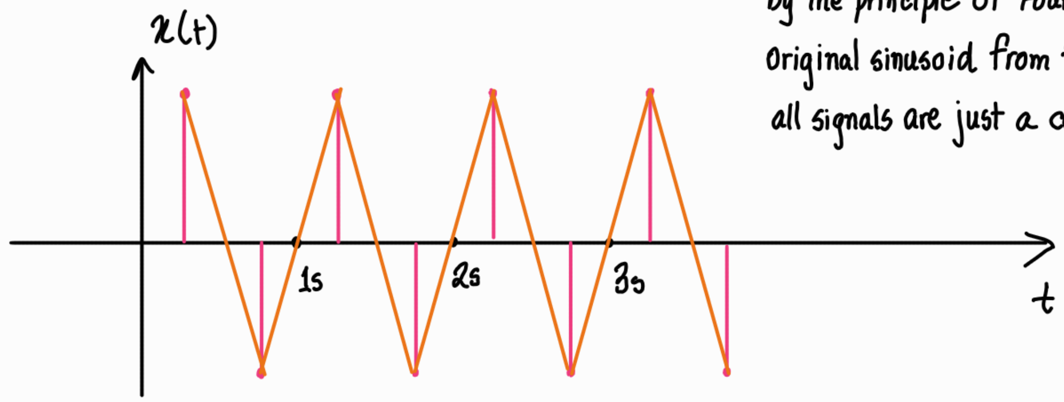


for $f_s = 2$ Hz (Nyquist frequency, f_N)



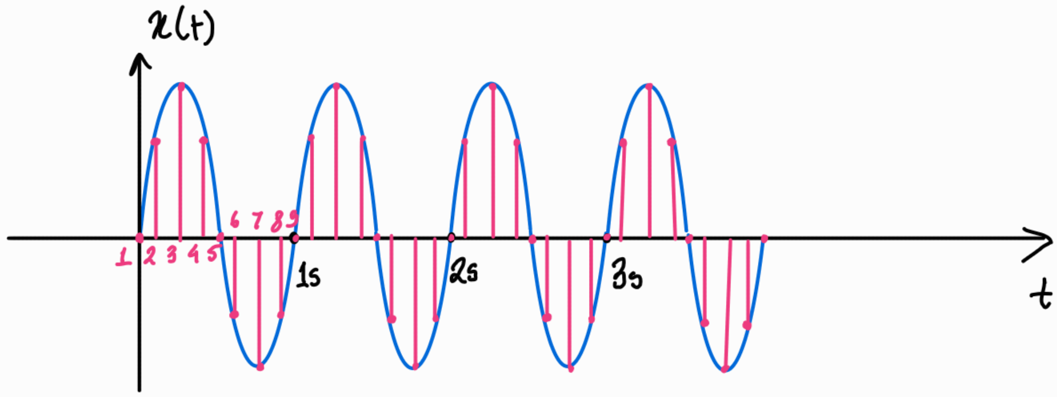
I know that I have offsetted the sampling train by 0.25s.

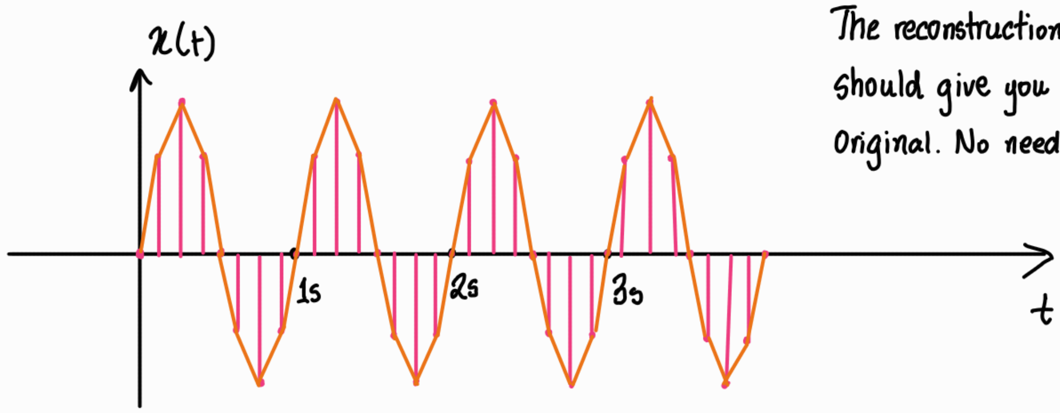
Reconstructing from DT —



at least the frequency is maintained. Going by the principle of Fourier, we still can get the original sinusoid from the signal, assuming all signals are just a combination of sinusoids.

If we increase the sampling frequency, reconstructions get easier. Assume $f_s = 9\text{Hz}$

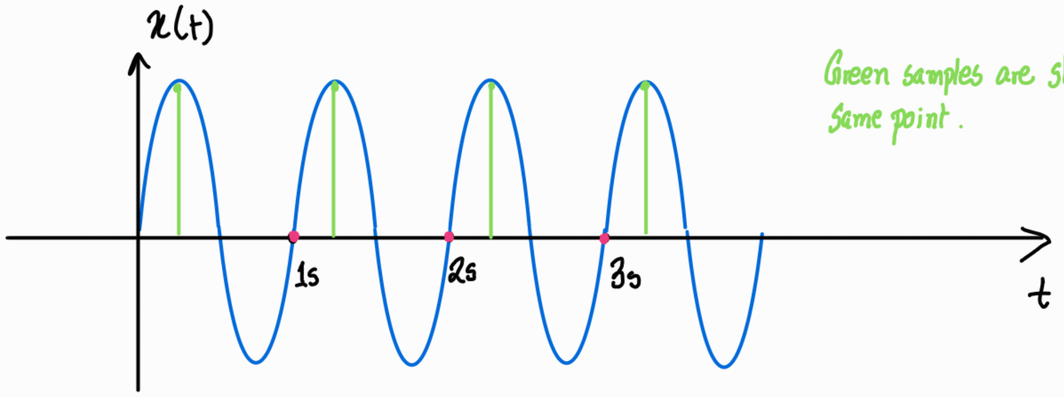




The reconstruction is way better. One LPF should give you a signal close to the original. No need to bother Fourier.

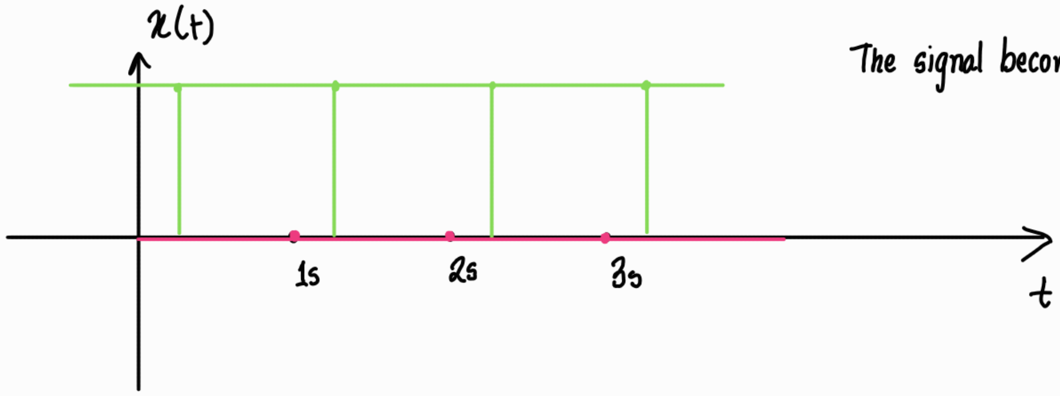
But what if $f_s < f_N$?

For $f_s = 1Hz$ —



Green samples are shifted a bit to prove the same point.

Reconstructing —



The signal becomes DC, loses all identity.

This is called aliasing. Why does this happen mathematically?

We know, $x[n] = A\sin\left(2\pi \frac{f}{f_s} n - \phi\right) = A\sin(\Omega n - \phi)$

In this case $x[n] = A\sin\left(2\pi \frac{1}{1} n\right)$ for no sampling offset.

then $x[n] = 0$ for any value of n as $n \in \mathbb{Z}$ and $\sin(2\pi n) = 0$

A deeper dive into normalized frequency —

For a given f_s, $max(f) = 0.5 f_s$ following Nyquist's theorem.

Now lets try out some f values and see how Ω changes —

for $f = -0.5 f_s \longrightarrow \Omega = -\pi$ rad/sample, $f_{norm} = -\frac{1}{2}$ cycles/sample; $\sin(-\pi n - \phi)$

for $f = 0$ Hz $\longrightarrow \Omega = 0$ rad/sample, $f_{norm} = 0$ cycles/sample ; $\sin(-\phi)$

for $f = 0.5 f_s \longrightarrow \Omega = \pi$ rad/sample, $f_{norm} = \frac{1}{2}$ cycles/sample ; $\sin(\pi n - \phi)$

Nothing interesting here. Something interesting happens when $f < -0.5 f_s$ or $f > 0.5 f_s$

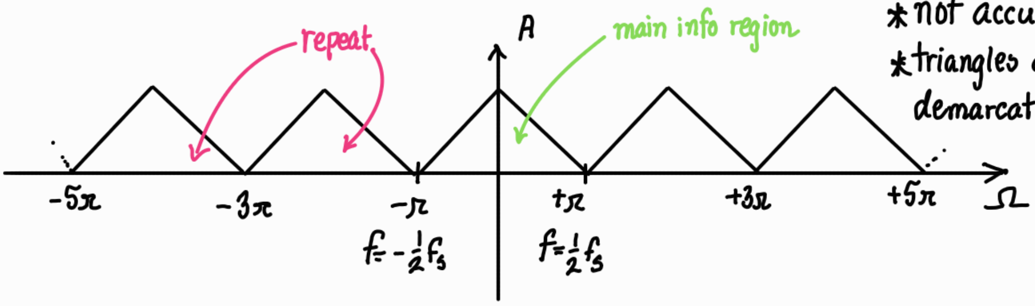
for $f = -f_s \longrightarrow x[n] = A\sin(-2\pi n - \phi) = A\sin(-\phi)$ implying $\Omega = 0 = -2\pi$ (rad/sample)

for $f = f_s \longrightarrow x[n] = A\sin(2\pi n - \phi) = A\sin(-\phi)$ implying $\Omega = 0 = 2\pi$ (rad/sample)

You can increase f anyhow you want, but the periodicty will always bring Ω to $[-\pi, \pi]$.

$\Omega = 2\pi$ rad/sample or $\Omega = -2\pi$ rad/sample are aliases of $\Omega = 0$ rad/sample. Or you can also say $f = -f_s$ or $f = +f_s$ components are aliases of $f = 0$ Hz when sampling under f_s. This means $\Omega = 2\pi$ or $\Omega = -2\pi$ doesn't hold any new information, rather its just a repeat of $\Omega = 0$. In fact, no new info exists outside $\Omega \in [-\pi, \pi]$

Now if there was a way to make Ω variable by aggregrating all contributions of n, we will probably get a graph like —



* not very apt premise
* not accurate graph
* triangles are used for demarcation

Question — Why waste so much time on sinusoids? Deriving normalized frequency limits for sine wave? Aren't real signals different from sinusoids?

Fourier states that all signal can be broken down into sinusoids. So everything we studied here are applicable for all signals because all signals are sinusoids.

Aperiodic DT sinusoids —

Most DT sinusoids are aperiodic. For $x[n] = A\sin(2\pi f_{norm} n - \phi)$ to be periodic there has to a N such that $x(n+N) = A\sin(2\pi f_{norm} n + 2\pi f_{norm} N - \phi)$

$$= A\sin(2\pi f_{norm} N + 2\pi f_{norm} n - \phi) = A\sin(2\pi f_{norm} n - \phi)$$

This is only possible iff $f_{norm} N \in \mathbb{Z}$. For $f_{norm} \cdot N \in \mathbb{Z}$, f_{norm} must be rational.

Fundamental period is $\min(N)$ for which $f_{norm} N \in \mathbb{Z}$. Fundamental frequency is then given by $\Omega_0 = \frac{2\pi}{N}$.

Example —

$x[n] = 10\sin(100n - 30°)$ is aperiodic

$x[n] = 10\sin\left(2\pi \cdot \frac{13}{37} n - 27°\right)$ is periodic $\longrightarrow$ $N = 37$, $\Omega_0 = \frac{2\pi}{37}$

Periodic DT sinusoids are not exactly similar to CT sinusoids.

Take this signal as an example — $x[n] = 10\sin\left(2\pi \cdot \frac{13}{37} n - 27°\right)$ Assume, $f = 13\text{Hz}$ & $f_s = 37\text{Hz}$

Here, CT signal's $f = 13$ Hz. So one cycle takes $\frac{1}{13}$s. But the DT signals cycle need 1s (N/f_s)

So the periodicity is not associated with the sine cycles. It is independent of the original sine.

Example —

Is $x[n] = 2\sin\left(\frac{5\pi}{7} n\right) + 5\sin\left(\frac{3\pi}{8} n\right)$ periodic and if so, what is the fundamental time period?

$\hookrightarrow f_1 = \frac{5}{14}$ $\quad\hookrightarrow f_2 = \frac{3}{16}$

$\because$ all components have rational normalized frequencies, $x[n]$ is periodic

here, $N_1 = 14$ & $N_2 = 16$. Then fundamental time period of $x[n]$ is given by $\text{LCM}(N_1, N_2) = 112$

Ending note — Time period of any aperiodic signal $= \infty$ while the cycle extends from $-\infty$ to $+\infty$